RAJALAKSHMI INSTITUTE OF TECHNOLOGY

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GE3451 - ENVIRONMENTAL SCIENCES AND SUSTAINABILITY

CLASS : II Year All Branches

UNIT V SUSTAINABILITY PRACTICES

Zero waste and R concept, Circular economy, ROHS, REACH 84, Material Life cycle assessment, Environmental Impact Assessment. Sustainable habitat: Green buildings, Green materials, Energy efficiency, Sustainable transports. Sustainable energy: Non-conventional Sources, Energy Cycles - carbon cycle, emission and sequestration, Green Engineering: Sustainable urbanization- Socio economical and technological change.

PREPARED BY **HOD** **PRINCIPAL**

Sustainability Practices

5.1. ZERO WASTE

Definition

Zero waste is a set of principles, focused on waste prevention, that encourages redesigning resource life cycles, so that all products are reused.

Goal

- ❖ The material should be reused until the optimum level of consumption is reached.
- ❖ It provides guidelines for continually working towards eliminating waste.
- ❖ To avoid sending trash to landfills, incinerators (or) the ocean.

Concept

The conservation of all the resources by means of responsible production, consumption, reuse and recovery of products, packaging and materials without burning and with no discharges to land, water (or) air that threaten the environment (or) human health.

Examples of zero waste

- ✓ one - way recyclable glass bottles.
- ✓ one - way milk bags.
- ✓ one - way aseptic cartons.
- ✓ one - way table - top paper board cartons.

Principles of zero waste

1. **Refuse what you don't need:**

- ➢ It prevents unwanted items from coming into your home.

2. **Reduce what you do use:**

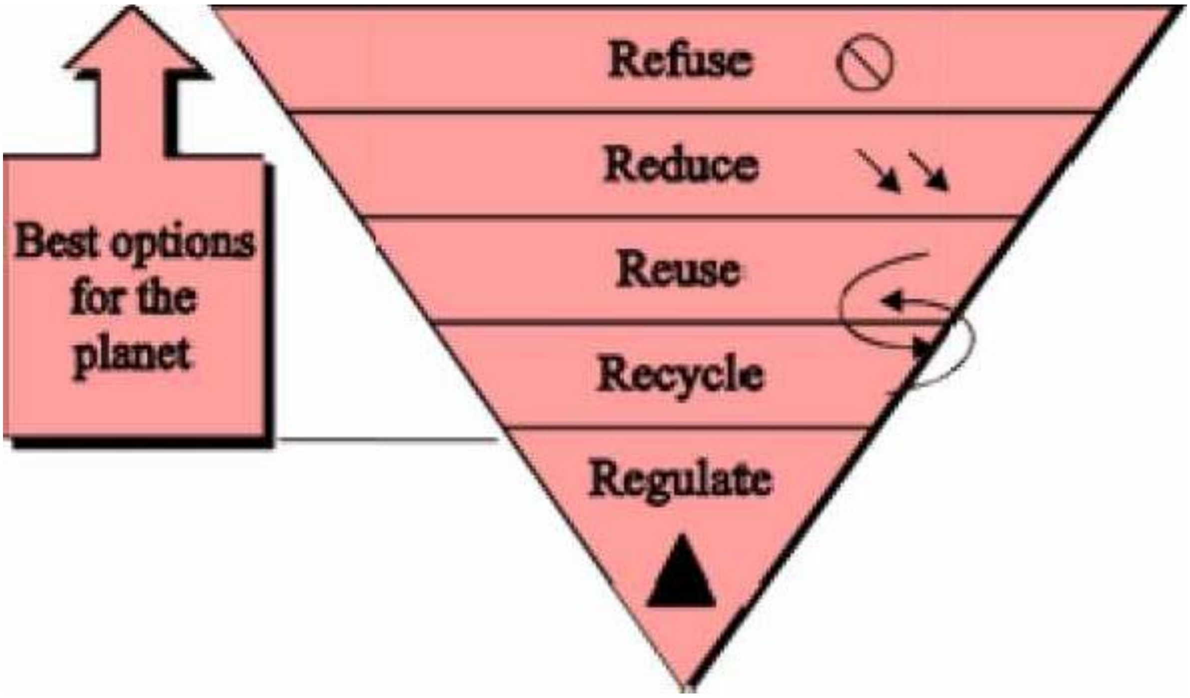
- ➢ It is equal to less waste at the end.

3. **Reuse whatever you can.**

4. **Recycle what you can't refuse (or) reduce.**

5. **Regulate of what's left over:**

Composting food scraps, paper pieces and wooden (or) bamboo tooth brushes returns nutrients and fiber back to the earth.



Zero waste Hierarchy

Steps to achieve zero waste

- Identify the high waste areas of our life-style.
- Know where to apply the principle of zero waste, if the waste cannot be removed (or) reduced.
- Substitute single use plastic with eco-friendly zero waste options.
- Buy zero waste (or) eco-friendly products.
- Support eco-friendly businesses.
- Put all your kitchen waste to good use (compositing).
- Reuse, up cycle and re-purpose.

Advantages and Disadvantages of zero waste

Advantages (or) Benefits

- ❖ Zero waste reduces our climate impact.
- ❖ It conserves resources and minimizes pollution.
- ❖ It promotes social equity and builds community.
- ❖ It supports a local circular economy and creates jobs.
- ❖ Zero waste needs businesses to play a key role.

Disadvantages (or) problem of zero waste

- ➢ Since zero wastes are solids, it is difficult to store.
- ➢ It is more expensive.
- ➢ Zero waste in time-consuming.
- ➢ It can cause anxiety.
- ➢ Zero waste can be misleading.
- ➢ It can be difficult for a large household.
- ➢ Zero waste products are hard to find.

5.2. R CONCEPT (OR) 3R CONCEPT (REDUCE,REUSE AND RECYCLE)

Definition

The principle of reducing waste, reusing and recycling resources and products is often called 3Rs.

I. Reduce

1. Reducing means choosing to use things with care to reduce the amount of waste generated.
2. If the usage of raw materials are reduced, the generation of waste also gets reduced.

II. Reuse

Reusing involves the repeated use of items (or) parts of items which still have usable aspects.

- ➢ The refillable containers, which are discarded after use, can be reused.
- ➢ Rubber rings can be made from the discarded cycle tubes, which reduces the waste generation during manufacturing of rubber bands.

III. Recycle

Recycling means the use of waste itself as the resources.

It involves reprocessing of the discarded materials into new useful products.

Examples

- ✓ Old aluminium cans and glass bottles are melted and recast into new cans and bottles.
- ✓ Preparation of cellulose insulation from paper.
- ✓ Preparation of fuel pellets from kitchen waste.
- ✓ Preparation of automobiles and construction materials from steel cans.

The above process saves money, energy, raw materials, and reduces pollution.

Concept of 3R

- ❖ The concepts of 3R refers to reduce, reuse and recycle, particularly in the topic of production and consumption.
- ❖ It forces for an increase in the ratio of recyclable materials, further reusing of raw materials and manufacturing wastes and overall reduction in resources and energy used.

Principle

3R is the order of priority of actions to be taken to reduce the amount of waste generated and to improve overall waste management processes and programs.

Importance of 3 Rs

- ❖ The most effective way to reduce the garbage is reducing the amount of solid waste produced.
- ❖ By reducing waste at the source, the resources like water and energy can be saved.
- ❖ Like reducing, reusing avoids creating waste rather than trying to recycle it once it’s already there.

- Operating a well-run recycling program costs less than waste collection and land filling.
- Recycling helps families save money because they pay for less disposal costs.
- Recycling produces less air and water pollution than manufacturing with new materials.
- By recycling less materials are sent to landfills, which will keep them for future.
- Proper disposal and recycling will prevent water and soil contamination.

Advantages (or) Benefits of 3 Rs

- ✓ Reduce greenhouse gas emissions.
- ✓ Saves energy.
- ✓ 3R is more energy consumption and pollution.
- ✓ 3R generates pollutants.
- ✓ Processing cost is high.
- ✓ Quality of resultant product is low.

5.3. CIRCULAR ECONOMY

Definition

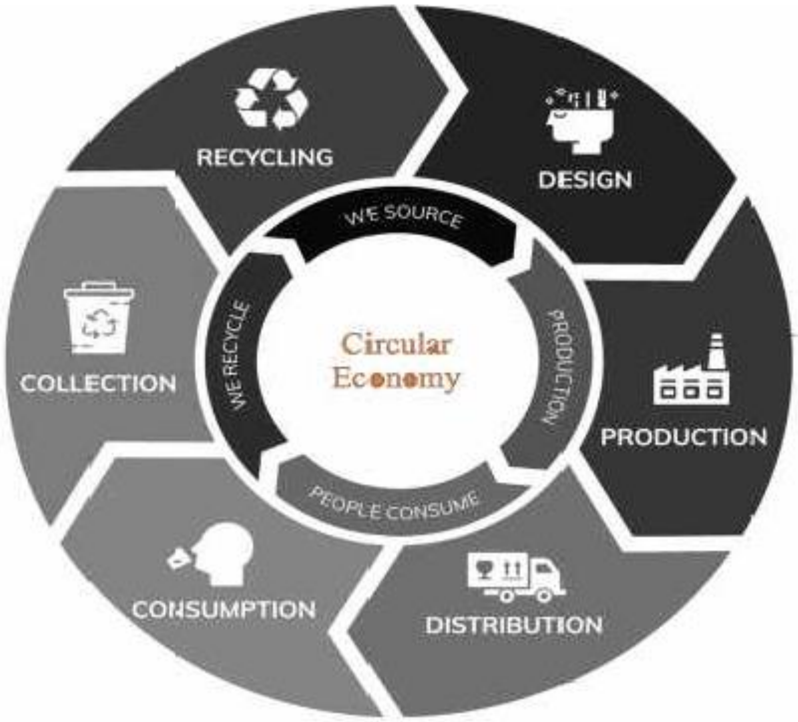
Circular economy ***is a new production and consumption model that ensures sustainable growth over time. It reduces the consumption of raw materials and recover wastes by recycling (or) giving it a second life as a new product.***

Aim (or) Purpose

Aim of the circular economy is to make the most of the material resources available to us by applying three basic principles reduce, reuse and recycle. In this way the life cycle of products is extended, waste is used and a more efficient and sustainable production model is established over time.

Benefits of circular economy

- ✓ It protects environment.
- ✓ Circular economy benefits the local economy.
- ✓ It drives employment growth.
- ✓ It promotes resource independence



Circular Economy

Necessary steps (7Rs) to achieve a circular economy

1. Redesign

Redesigning process consumes fewer raw materials, extends their life cycle and generates less waste.

2. Reduce

If we reduce consumption, waste generation and use of raw materials, impact on the environment gets reduced.

3. Reuse

Reusing the products extends their life cycle.

4. Repair

Repairing avoids the use of new raw materials, saves energy and does not generate environmental waste.

5. Renovate

Update old objects, so that they can be reused.

6. Recycle

Waste product can be used as raw material to manufacture new products.

7. Recover

The products that are going to be discarded, can be used for new uses.

Example for Circular Economy

- ✓ Manufacturers design products to be reusable.
- ✓ Electrical devices are designed in such a way that they are easier to repair. Products and raw materials are also reused as much as possible.

5.4. RoHS and REACH

5.4.1. RoHS

RoHS (Restriction of Hazardous Substances) and REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) are two environmental regulations for hazardous material. They apply to companies selling products in the European Union (EU).

RoHS (**R**estriction **O**f **H**azardous **S**ubstances)

- RoHS stands for Restriction of Hazardous Substances, which is a European Union directive that **limits the amount of hazardous chemicals in electrical and electronic equipment (EEE).**
- The directive was adopted in February 2003 to protect public health and the environment.
- RoHS aims to make electronic equipment safer and more environmentally friendly by limiting the use of hazardous substances like lead, mercury, and certain flame retardants.
- It applies to a wide range of EEE, including appliances, consumer electronics, and industrial equipment.
- RoHS compliance is mandatory in many countries, including the EU, UAE, China, Taiwan, Singapore, Japan, Turkey, South Korea, Saudi Arabia, India, and more.
- Non-compliance can result in legal penalties.

RoHS is often referred to as the "lead-free directive", but it restricts the use of the following ten substances:

1. Lead (Pb)
2. Mercury (Hg)
3. Cadmium (Cd)
4. Hexavalent chromium (Cr6+)
5. Polybrominated biphenyls (PBB)
6. Polybrominated diphenyl ether (PBDE)
7. Bis(2-ethylhexyl) phthalate (DEHP)
8. Butyl benzyl phthalate (BBP)
9. Dibutyl phthalate (DBP)
10. Diisobutyl phthalate (DIBP)

Maximum Permitted Concentration of all the above chemicals are 0.1% except Cadmium (max - 0.01%)

5.4.2. REACH (**R**egistration, **E**valuation, **A**uthorisation and Restriction of **Ch**emicals)

REACH is a European Union regulation that **protects human health and the environment from the risks of chemicals.** It came into effect on June 1, 2007.

Purpose: REACH ensures that **chemicals are produced and used safely by requiring companies to identify and manage risks**. It also aims to **enhance the competitiveness and innovation of the EU chemicals industry**.

How it works: Companies must register information on their chemical substances with the European Chemicals Agency (ECHA). The ECHA then evaluates the information and coordinates the phasing out or restriction of substances of high concern.

Who it applies to: REACH applies to all chemical substances and related businesses.

Where to find the regulation: The consolidated version of the REACH regulation is available in EUR-lex, the online portal to EU law.

The REACH Regulation includes a number of measures, such as:

1. Authorization

A licensing system that prevents certain substances from being used or placed on the market in the EU unless the company has authorization.

2. Restrictions

Conditions that prohibit or limit the manufacture, use, or placing on the market of a substance. Restrictions can include limiting the use of a substance in certain products or prohibiting its use altogether.

REACH registration number

An 18-digit number assigned by the European Chemicals Agency (ECHA) that verifies a company has completed their registration for a chemical substance.

5.5. MATERIAL LIFE CYCLE ASSESMENT

Definition

Life cycle assessment (LCA) ***is a process of evaluating the effects of a material on the environment over the entire period of its life, there by increasing resource use efficiency and decreasing liabilities.***

Generally LCA is used to study the environmental impact of a material.

LCA is commonly referred to as a cradle-to-grave analysis

Stages of a life cycle assessment

The followings are the 5 stages of a life cycle assessment

Step 1: Raw materials (Resources) extraction and processing.

Step 2: Manufacturing.

Step 3: Transportation.

Step 4: Distribution.

Step 5: Usage and retail.

Step 6: Waste disposal (end of life).

Life cycle assessment

In the manufactured product, environmental impacts are assessed from raw material extraction and processing, through the product's manufacture, distribution and use, to the recycling (or) final disposal of the materials.

Benefits (or) Advantages of LCA

- ❑ LCA is widely used to support sustainable development.
- ❑ LCA allows decision makers to compare two products and to select the product that has lowest impact on the environment.
- ❑ It is a modelling tool to assess environmental impacts of a product during its entire lifespan.
- ❑ LCA provides a holistic view on the environmental impacts, to avoid optimizing one environmental indicator without considering the effects on the other indicators.
- ❑ LCA identifies hotspots in the environmental impact.
- ❑ LCA is purely based on internationally accepted standards.

Disadvantages (or) Limitations

- ❖ LCA assesses the real world in a simplified model.
- ❖ The assumptions, scenaries and scope may vary from one study to the other leading to different LCA results.
- ❖ Variations in LCA approaches and results may be confusing especially for non-experts.
- ❖ LCA study requires large amount of data.
- ❖ If data collection is poor, the study will not lead to solid conclusions.

- It is not easy to communicate the results of a LCA study.

5.6. ENVIRONMENTAL IMPACT ASSESSMENT

- EIA is defined as a formal process of predicting the environmental consequences of any development projects.
- It is used to identify the environmental, social and economic impacts of the project prior to decision making.

Purpose (or) Aim of EIA

- The main purpose of EIA is to determine the potential environmental, social and health effects of a proposed developmental projects.

Objectives of EIA

To identify the main issues and problem of the parties.

To identify who is the party.

To identify what are the problems of the parties.

To identify why the problems are arise.

Benefits of EIA

- ✓ Cost and time of the project is reduced.
- ✓ Performance of the project is improved.
- ✓ Waste treatment and cleaning expenses are minimised.
- ✓ Usages of resources are decreased.
- ✓ Biodiversity is maintained.
- ✓ Human health is improved.
- ✓ It helps in preventing natural calamities like earthquake, cyclone, etc.,

Process of EIA (or) Key Elements of EIA

- The key elements used in the process of EIA are
    - ✓ Scoping
    - ✓ Screening
    - ✓ Identifying and evaluating alternatives
    - ✓ Mitigating measures dealing with uncertainty
    - ✓ Issuing environmental statements

1. Scoping

- It is used to identify the key issues of the concern in the planning process at an early stage. It is also used to aid site selection and identify any possible alternatives.

2. Screening

- It is used to decide whether an EIA is required (or) not based on the information collected.

3. Identifying and evaluating alternatives

- It involves knowing alternative sites and alternative techniques and their impacts.

4. Mitigating measures dealing with uncertainty

- It reviews the action taken to prevent (or) minimize the adverse effects of a project.

5. Environmental statements

- This is the final stage of the EIA process. It reports the findings of the EIA.

5.7. SUSTAINABLE HABITAT

- Sustainable habitat means the maintenance of our natural home.

Definition

A sustainable habitat is an ecosystem that produces food and shelter for people and other organisms without resource depletion ie., no external waste is produced.

Features (or) Characteristics of sustainable habitat

- Proper waste management.
- Affordable housing.
- Waste water treatment and facility of recycling waste water.
- Green transportation using green fuel like biodiesel.

Objectives of national mission on sustainable habitat

- To reduce energy demand by promoting alternative technologies and energy conservation practices in both residential and commercial areas.
- Better urban planning like
    - ✓ using better disaster management
    - ✓ lesser use of private transport
    - ✓ more usage of public transport
- Encourage community involvement and participation of stake holders.

- ❖ Conservation of natural resources such as clean air, water, flora and fauna.
- ❖ Facilitate the growth of small and medium cities.
- ❖ To create sustainable habitats, engineers and architects should not consider any element as a waste product.

How to maintain sustainable habitat

- ❑ For maintaining our sustainable habitat, we should
  - ✓ Promote energy efficiency.
  - ✓ Promote the use of eco-friendly fuels.
  - ✓ Better manage municipal solid waste.
  - ✓ Promote to public transport.

5.8. GREEN BUILDINGS

Definition

Green building is an efficient method of construction that produces healthier buildings, which have less impact on the environment and climate. It requires less cost to maintain.

Green buildings preserve previous natural resources and improve our quality of life.

Criteria for green building

1. Green builders are encouraged to build on previously developed land rather than developing new land.
2. It is also important to build near existing infrastructure like bus routes, market, libraries.
3. The building site should be smaller because there is less environmental foot print.
4. Sites must be sustainably landscaped and don't suffer from soil erosion (or) light pollution.
5. Water reduction is built in by design using low-flow toilets, grey water systems.
6. Green buildings are constructed using clean energy like geothermal, solar, wind energies.
7. Green builders reduce material usage wherever possible. Mainly they use natural, renewable sources.
8. Selecting low emitting materials and products not only improves human health but also protect the overall environment.

Features of green building

- Efficient use of energy, water and other resources.
- Use of renewable energy such as solar energy.
- Pollution and waste reduction measures ie., reuse and recycling.
- Good indoor environmental air quality.
- Use of materials that are non-toxic, ethical and sustainable.
- A design that enables adaptation to a changing environment.
- Consideration of the quality of life of occupants in design, construction and operation.
- Construction of the environment in design, construction and operation.

Thus, any building can be a green building whether it is a home, an office, a school, a hospital, a community centre provided it includes features listed above.

Principles of green building

- The five principles of green building are
  - ✓ Livable communities.
  - ✓ Energy efficiency.
  - ✓ Indoor air quality.
  - ✓ Resource conservation.
  - ✓ Water conservation.

Components of green building

Seven important components of green buildings are

- ✓ Aluminium weather resistant insulated access panel. It helps regulate in door temperature and prevent moisture and pest from entering.
- ✓ Energy efficient windows.
- ✓ Green roof.
- ✓ Solar power.
- ✓ Water conservation.
- ✓ Recycling.
- ✓ Landscaping.

Advantages and Disadvantages of green building

Advantages of green buildings

- ✓ Green buildings are energy efficient.
- ✓ Higher fraction of eco-friendly materials.
- ✓ Water - efficient devices.
- ✓ Reduction in waste.
- ✓ Less air pollution.
- ✓ Reduction in green house gas emissions.
- ✓ Protection of our natural resources.
- ✓ Indoor air quality is improved.
- ✓ Use of recycled metal and other construction materials.
- ✓ Emphasis on renewable energies.
- ✓ Day lighting is utilized as best as possible.
- ✓ Use of renewable plant materials.
- ✓ Higher market value.
- ✓ Rainwater collection and use of compost bins.
- ✓ Overall health improvements.

Disadvantages of green building

- ✓ High initial costs.
- ✓ Energy supply may depend on weather condition.
- ✓ Technology problems are more.
- ✓ Maintenance may be difficult.
- ✓ Indoor air temperature may greatly vary over time.
- ✓ Experienced green construction workers may be rare.
- ✓ Green construction is not suitable for all locations.
- ✓ Availability of green construction materials.
- ✓ Funding problems for green buildings.

5.9. GREEN MATERIALS

Definition

Green materials also called eco-friendly materials, building construction materials that have low impact on the environment. Due to the properties of non-toxic, organic and recycling, green materials are widely used in various industrial applications.

Examples: Naturally occurring materials like wood, ceramics, glass, clay, sand, stone.

Criteria for green materials

Following criteria can be used to identify the green materials.

- ✓ Local availability of materials.
- ✓ Embodied energy of materials.
- ✓ % of recycled (or) waste materials used.
- ✓ Rapidly renewable materials.
- ✓ Contribution in energy efficiency of building.
- ✓ Recyclability of materials.
- ✓ Durability.
- ✓ Environmental impact.

Characteristics of green materials

Common characteristics of green materials are

- ❖ Green materials are energy efficient products, it uses less energy to do the same task.
- ❖ It lowers energy cost and lessen pollution.
- ❖ Green materials are mostly renewable, can be regenerated again and again.

Example : Bamboo grows quickly while pine grows **more slowly, but both are renewable.**

- ❖ Green materials are recyclable (or) made from recycled material. So, they save energy and reduce waste.
- ❖ Green materials are non-toxic, they do not emit odors, irritants (or) hazardous compounds that affect human health.
- ❖ They are durable and no need to upgrade (or) repair. They preserve resources and energy.
- ❖ They are cost-effective.
- ❖ They can be locally sourced, so transport cost can be reduced.

Important green building materials

Green building is construction that primarily uses natural materials and renewable resources. These structures look really cool.

1. **Stone: It is low maintenance and durable.**

2. Cob:

(mud mixture of natural ingredients like soil, sand, straw and lime).
It is cheap and energy efficient.

3. Bamboo

It is durable and light weight.

4. Cork: (Cork canes from oak trees).

It is a very good thermal insulator and mold resistant

5. Adobe brick: (brick made of clay and straw).

Natural noise protection and posses unique design (can be easily cut and transformed).

6. Straw bale

Easily renewable and cheap.

7. Cord wood

Affordable (cheap and easy construction), thermal efficiency.

8. Earth bags (or) sand bags

Locally sourced and provide natural insulation.

9. Mycelium (or) mushroom roots

Strong and light weight.

Examples of green materials

- ✓ Bamboo floorings.
- ✓ LED lightings.
- ✓ Reclaimed wood.
- ✓ Energy efficient appliances.
- ✓ High-efficiency glass windows.
- ✓ Solar panels.
- ✓ Recycled steel.
- ✓ Cork.
- ✓ Precast concrete slabs.
- ✓ Low VOC paint.

5.10. ENERGY EFFICIENCY

Definition

Energy efficiency is the use of less energy to perform the same task (or) produce the same result.

Energy efficient homes and buildings use less energy to heat, cool and run appliances and electronics.

Energy efficiency logo

Methods of achieving energy efficiency

Energy efficiency can be achieved by the following methods.

- Alternative waste treatment.
- Avoided emissions from diverting legacy waste from landfill for process engineered fuel manufacture.
- Avoided emissions from diverting legacy waste from landfill through a composting alternative waste technology.
- Capture and combustion of landfill gas.

Calculation of energy efficiency

- Energy efficiency can be calculated using the following relation.

$$\text{Energy efficiency} = \frac{\text{energy output}}{\text{energy input}} \times 100\%$$

Advantages (or) Benefits of energy efficiency

- Using energy more efficiently is one of the fastest, most cost - effective ways to save money.

- Increased energy efficiency can lower greenhouse gas emissions and other pollutants.
- Energy efficiency also decreases water use.
- It can lower individual utility bills, create jobs and help stabilize electricity prices.
- It provides long-term benefits by lowering overall electricity demand, thus reducing the need to invest in new electricity generation and transmission infrastructure.
- Energy efficient construction is environmentally - friendly as it does not emit harmful carbon dioxide into the atmosphere.

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5.11. SUSTAINABLE TRANSPORT

Definition

Sustainable transport refers to ***any means of transportation that is "green" and has low impact on the environment.***

Examples

- ✓ walking
- ✓ cycling
- ✓ transit
- ✓ carpooling
- ✓ car sharing
- ✓ green vehicles

- Sustainable transport can carry people for more efficiently than cars.
- Electric cars pollute less and reduce individual carbon foot prints.

Sustainable transport

Importance of sustainable transport

- Sustainable transport contributes to reduction in damaging CO emission and therefore to a reduction in atmospheric pollution and improved air quality in cities.
- The aim of this type of transport is to reduce the negative impacts on the environment.

Key elements of sustainable transport

1. Fuel economy

- The better fuel economy gets the lower emissions go. By improving fuel economy we can get the same mileage while generating fewer emissions.
- It is achieved by
  - ✓ making engines more efficient.

2. Occupancy

The cheapest and simplest way to lower the carbon intensity of a vehicle is to stick more people in the vehicle.

Example

- ✓ Local bus has emissions 7 times higher than the school bus.
- ✓ The main difference is that the school bus has very high occupancy.

3. Electrification

Electrification is the most important pathway to low carbon transport.

4. Pedal power

Bicycles reduces the carbon emissions.

5. Urbanization

It is a huge opportunity for lowering both distance travelled per person and the carbon intensity of that travel.

How to Promote sustainable transport

Followings are steps for promoting sustainable transport.

1. ***Enhancing public transportation:***

❖ It is not only less polluting means of transportation, but also promoting HSE (Health, safety and environment) policy.

2. ***Encouraging car pooling:***

❖ It reduces the volume of CO , emitted per inhabitant.

3. ***Encouraging bicycle use:***

❖ It is reliable and non-polluting means of transportation.

4. ***Teleworking:***

❖ It reduces employee travel and therefore their carbon food print.

5. ***Improving the parking experience:***

❖ It can be done effectively with the help of a parking management

software.

Advantages and Disadvantages of sustainable transport

Advantages (or) benefits

- ✓ It creates job.
- ✓ Provides safer transportation.
- ✓ Emits less pollution.
- ✓ Promotes health (sustainable transit reduces emissions and air pollution)
- ✓ It saves energy.
- ✓ Saves money.

7. ***Decreases congestion:***

❑ When people choose sustainable transportation, over driving themselves, congestion also decreases.

8. ***It conserves land.***

❑ It encourages compact development, fewer roadways in country areas results in less runoff, thereby protecting the land and the biodiversity.

Disadvantages (or) limitations

- ✓ Modifications to handling and transport facilities.
- ✓ The initial purchase of reusable containers.
- ✓ Additional costs of the tracking system e.g., software packages, reading equipments, electronic chips, barcode labelling, detector's etc.,

5.12. SUSTAINABLE ENERGY

Definition

Sustainable energy is the energy which meets the needs of present without compromising the ability of future generations to meet their own needs.

- ❑ It should be encouraged as it does not cause any harm to the environment and is available widely at free of cost.

Sources of sustainable energy

- ❖ Followings are the sustainable enery sources as they are stable and available in plenty.
  - ✓ Wind energy.
  - ✓ Solar energy.
  - ✓ Ocean energy.
  - ✓ Hydro power.
  - ✓ Geothermal energy.

Advantages and disadvantages of sustainable energy

Advantages (or) Benefits

1. Improves public health

- ❑ Burning of fossil fuels produces serious public health issues like neurological damage, cancer, heart attacks, breathing problems and premature death.
- ❑ However these problems can be eliminated by using sustainable energy sources, which emit no air (or) water pollutants.

2. Creates local jobs

- ❑ Since most of the sustainable energy infrastructure is built locally (or) in the same country, it helps creates jobs and improves the economy.

3. Decrease your carbon footprint

- ❑ Sustainable energy like wind and solar energy creates zero carbon emissions.

4. Cost saving

- As it is easily available they are much more cost-effective than traditional energy resources, such as power plants

5. Energy security

- It helps to conserve the planet's natural resources and reduce the pollution.

Disadvantages (or) limitations

- Sustainable energy sources are not available round the clock.
- The efficiency of sustainable energy technologies is low.
- The initial cost of sustainable energy is high.
- Sustainable energy sites require a lot of space.
- Sustainable energy devices need recycling.

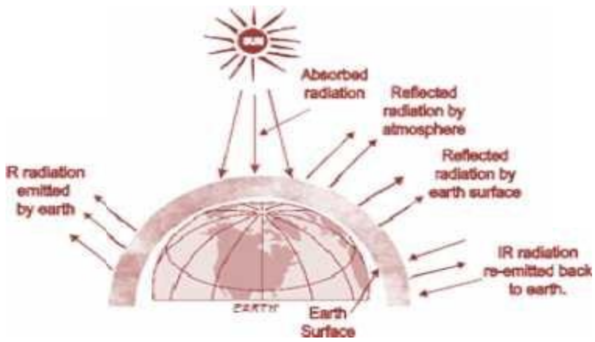
5.13. ENERGY CYCLE

Energy cycle, is the interactions between energy sources within the Earth's environment.

- These interactions are very complex and even small changes in them can lead to significant changes in long-term climate behavior.

Illustration

- A simple illustration of the major elements of the energy cycle is shown in the figure. 5.16 Soil moisture is an important factor in the absorption and reflection of the sun's energy by the earth's surface.



Energy Cycles

- Important energy cycles
  - ✓ Carbon cycle.
  - ✓ Nitrogen cycle.
  - ✓ Phosphorus cycle.

CARBON CYCLE

Definition

Carbon cycle is the movement of carbon (or) carbon compounds continuously from the atmosphere to the earth and then back into the atmosphere.

(Or)

Carbon cycle is the process where carbon compounds are interchanged among the biosphere, geosphere, hydrosphere and atmosphere of the earth.

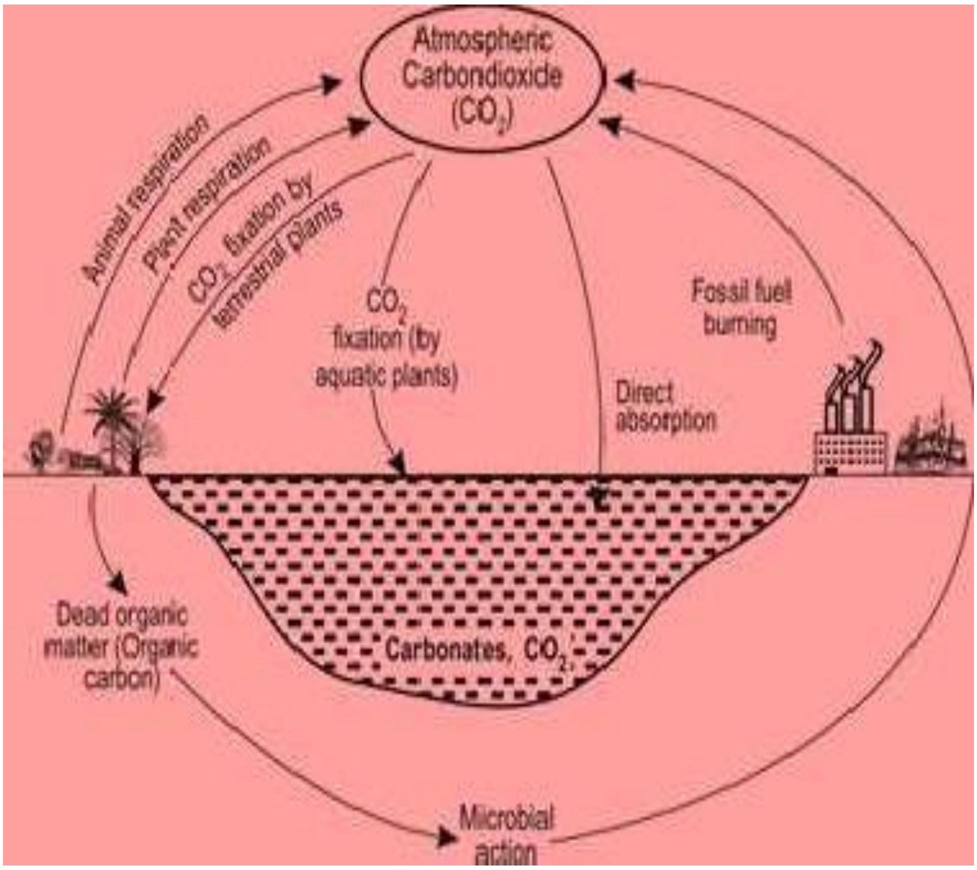
- Carbon in the atmosphere is present in the form of carbon dioxide.
- Carbon enters the atmosphere through natural process such as respiration and industrial applications such as burning of fossil fuels

Sources of CO in atmosphere

- During respiration, plants and animals liberates CO in the atmosphere.
- Combustion of fuels also release CO
- Volcanic eruptions also release CO .

Various steps involved in carbon cycle

- Carbon cycle involves the following 5 important steps.



Carbon cycle

Step I:

- Carbon present in the atmosphere is absorbed by plants by the processes photosynthesis, which involves the absorption of CO by plants to produce carbohydrates (producers).

$$CO_2 + H_2O + energy \text{ -----> } (CH_2O)_n + O_2$$

Step II:

- These plants are then consumed by animals and carbon gets bioaccumulated into their bodies (consumers).

Step III:

- These animals and plants eventually die and decomposers eat the dead organism and return the carbon from their body back into the atmosphere (decomposers)

$$(CH_2O) + O_2 \text{ -----> } CO_2 + H_2O + energy$$

Step IV:

- Some of the carbon that is not released back into the atmosphere eventually become fossil fuels.

Step V:

- These fossil fuels are then used for man-made activities, which pump more carbon back into the atmosphere.

Importance (or) benefits of carbon cycle

- It plays a vital role in balancing the energy and traps the long-wave radiations from the sun ie., it acts like a blanket over the planet, avoids global warming.
- Carbon cycle is an important aspect of the survival of all life on earth.
- Carbon is the building block of life and forms bonds with other elements necessary for life.

5.15. CARBON EMISSION AND SEQUESTRATION

Carbon emission

Carbon emission is the release of green house gases and their precursors into the atmosphere over a specified area and period of time.

Types of carbon emission

- Carbon (Green house gas) emissions are classified into two scopes.

1 Scope 1 emissions (or) Direct emission

Scope 1 emissions are direct emissions from company. It is divided into 4 categories.

(a) Stationary combustion

- All fuels that produce GHG.

(b) Mobile combustion

- All vehicles owned by a firm, burning fuel.

  Example: cars, vans, trucks.

(c) Fugitive emissions

- These are leaks from green house gases (GHG).

  Example: Refrigeration, air-conditioning units.

(d) Process emissions

- These are from industrial processes and on-site manufacturing.

Example: Cement manufacturing, chemical manufacturing.

II Scope 2 emissions (or) Indirect emission

- Scope 2 emissions are indirect emissions from the generation of purchased energy (purchased electricity steam, heat and cooling) from a utility provider (end user).

Sources (or) Causes of carbon emission

1. Natural sources of CO emission

- It includes
  - ✓ Decomposition of matter.
  - ✓ Ocean release.
  - ✓ Respiration.
  - ✓ Most animals, which exhale CO as a waste product.
  - ✓ Carbonate rocks.
  - ✓

2. Human sources of CO_2 emission

- It includes
  - ✓ Burning of fossil fuels like coal, natural gas and oil.
  - ✓ Deforestation.
  - ✓ Industrial activities like cement manufacture, oil refineries and leather industries.
  - ✓ Transportation sector generates largest amount of CO_2 in the atmosphere.

Harmful effect of carbon emission

- Carbon emission, nothing but emission of green house gas, affects the planet significantly.
- It causes global warming and affects climate change.

Reduction of carbon emission

- There are many ways to reduce green house gas emissions like
  - ✓ energy efficiency.
  - ✓ fuel switching.
  - ✓ combined heat and power.
  - ✓ use of renewable energy.
  - ✓ more efficient use.
  - ✓ recycling of materials.
  - ✓ plant more trees.
  - ✓ reduce air travel.
  - ✓ driving more efficient.

Carbon sequestration

- It is the process of capturing and storing atmospheric carbondioxide.
- It is one method of reducing the amount of CO_2 in the atmosphere.
- Goal of carbon sequestration is to reduce global climate change.
- 25% of our carbon emissions have been captured by earth's forests, farms and grassland.
- Scientists and land managers are working to keep landscapes vegetated and soil hydrated for plants to grow and sequester carbon.

- ❖ 30% of the carbon dioxide, we emit from burning fossil fuels, is absorbed by the upper layer of the ocean.
- ❖ 45% of carbon dioxide stays in the atmosphere the rest in sequestered naturally by the environment.

Concept (or) Aim of carbon sequestration

- ❑ The concept of carbon sequestration is to stabilize carbon in solid and dissolved forms so that it doesn't cause the atmosphere to warm.
- ❑ The process shows tremendous promise for reducing the human "carbon foot print".

Method (or) Types of carbon sequestration

- ❑ There are three main types of carbon sequestration.

1. Biological carbon sequestration

- ❖ It is the storage of CO in vegetation like grassland, forests, soils and oceans.

2. Geological carbon sequestration

- ❖ It is the process of storing CO in underground geologic formations (or) rocks.
- ❖ Typically, CO is captured from an industrial sources like steel (or) cement production, power plant and injected into the porous rocks for long-term storage.

3. Technological carbon sequestration

- ❖ Scientists are using innovative technologies to remove and store carbon from the atmosphere using innovative technologies.

Example: Graphene production

- ❑ The use of CO as a raw material to produce graphene (a technological material).
- ❑ Graphene is used to create screens for smart phones and other technical devices.
- ❑ Graphene production is an example of how CO can be used as a resource and a solution in reducing emissions from atmosphere.

Advantages and disadvantages of carbon sequestration

Advantages (or) merits

- ✓ Carbon sequestration prevents the occurrence of climate change.
- ✓ Deep injection of CO improves the extraction of fuels like oil and methane from their reserves in addition to removing excess pollutants from the air.
- ✓ Since the gas can be easily liquefied, it can be easily transmitted through pipelines.

- ✓ No CO leaking out from the injection site.
- ✓ It lowers carbon emission by 80% to 85% while using fossil fuels.

Disadvantages (or) limitations

- ✓ Due to carbon sequestration, in power plants, 40% additional coal is consumed and hence cost of energy gets increased by 1 to 5% per kilowatt hour.
- ✓ CO from power plant emissions must be captured and liquified, which uses a lot of electrical power.
- ✓ It can be disastrous if the injected gas leaks due to structural flaws in the geological formation.
- ✓ The ocean can become acidic due to the large amounts of carbon dioxide being injected into it, endangering aquatic life.
- ✓ Planting trees, with the intention of storing and absorbing carbon, requires more time for the trees to mature.
- ✓ There is no enough available geological resources to sequester carbon.
- ✓ The concentration of CO_2, from power plant exhaust is too low for being effectively liquified.

5.16. GREEN ENGINEERING

Definition

Green engineering is the design, commercialization and use of processes and products that minimizes pollution, promotes sustainability and promotes human health without affecting environment.

Examples for green engineering

- ➢ Biodegradable cups and straws.
- ➢ Enhanced industrial emission filters.
- ➢ Waste water treatment.
- ➢ Radiant floors (heat homes efficiently by installing warming tubes under a floor).
- ➢ Plant-based cooling (an alternate cooling solution using plants and trees installed around (or) on a building)

Goal of green engineering

- Decrease in the amount of pollution that is generated by a construction.
- Minimization of human population exposure to potential hazards (reducing toxicity).
- Improved uses of matter and energy throughout the life cycle of the product.
- Maintaining economic efficiency and viability.
- Reduces energy and water consumption.
- Reduces waste and our carbon footprint.
- Improves business efficiency by lowering costs while improving the product design and creating new jobs.

Principles of green engineering

- All materials and energy inputs and outputs are inherently non-hazardous as possible.
- It is better to prevent waste that to treat (or) clean up waste after it is formed.
- Separation and purification operations should be designed to minimize energy consumption and material use.
- Products, processes and systems must be designed to maximize mass, energy, space and time efficiency.
- Products, processes and system should be “output pulled” rather than “input pushed” through the use of energy and materials.
- Complexity must be viewed as an investment when making design choices on recycle, reuse.
- Durability rather than immortality should be a design goal.
- Material diversity in multi-component products should be minimized.
- Design of products, processes and system must include integration and inter-connectivity with available energy and materials flow.
- Products should be designed for performance in a commercial “after life”.
- Material and energy inputs should be renewable rather than depleting.

Benefits of green engineering

- This process enhances business practices by eliminating improper production methods.
- It improves a company’s reputation by showing consumers it cares about the environment.

- It minimizes energy (or) production waste.
- It provides tax incentives.
- It helps the global environment.
- It reduces air, water and soil pollutions.
- It provides new business opportunities.

Limitations (or) disadvantages of green engineering

- R & D costs, production and implementation costs are high.
- Implementation will take many years.
- Green technology is still quite immature.
- Some companies may go out of business.
- Job losses.
- Sophisticated regulatory frame work needed.
- Not everything that is labeled as green is actually green.

5.17. SUSTAINABLE URBANIZATION

- Urbanization is the movement of human population from rural areas to urban areas for the want of better education, communication, health, employment, etc., without affecting the environment and needs of future generations.

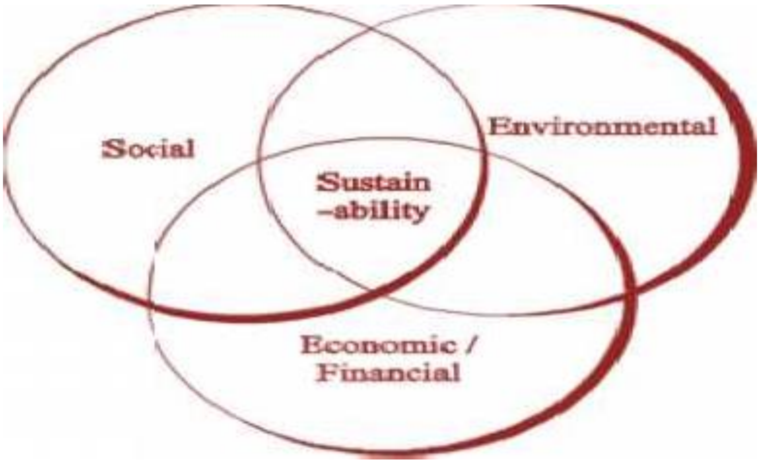
Rules to develop a sustainable urbanization

- Sustainable transportation.
- Sustainable urban development.
- Climate change mitigation and landscape architecture.
- Resilient design (regarding natural hazards).
- Applying ecological design.
- Improving water efficiency.
- Increasing energy efficiency.
- Using low-impact materials.
- By following the above rules, urbanization can be made into sustainable.

Pillars of sustainable urbanization

- Sustainability is based on three functional areas ie., social, environmental and financial/economical.

- These functional areas are interconnected and must be considered together.
- The place where these all meet and are balanced is the goal of sustainability.



Functional areas of urban sustainability

- The goal of urban sustainability is to prevent resource availability issues for existing (or) future generations.
- It also minimizes an urban area's impact on its ecosystem.

Advantages and disadvantages of sustainable urbanization

Advantages

- ✓ Urbanization creates convenience.
- ✓ Urban economies can be better than rural ones.
- ✓ Provides better education.
- ✓ Get better housing.
- ✓ Provides better social life.
- ✓ Provides better healthcare services.
- ✓ More security and police availability.
- ✓ More entertainment options.
- ✓ More tourist attractions.
- ✓ More places to shop in urban areas.

Disadvantages

- Over crowding in urban areas.
- Buying a house might be a challenge.
- Decline in rural area.
- Too much crime occurs in urban area.

- ❖ Unemployment problem is more.
- ❖ Cost of living is higher.
- ❖ No privacy.
- ❖ Pollution problem is more.

5.18. SOCIO – ECONOMICAL CHANGE ON SUSTAINABLE URBANIZATION

- ❑ Urbanization has many adverse effects on the structure of society because,
  - ➢ gigantic concentrations of people compete for limited resources.
  - ➢ rapid housing construction leads to overcrowding.
  - ➢ slums, which experience major problems such as poverty, poor sanitation, unemployment.
  - ➢ it leads to higher crime rates and pollution.
  - ➢ it also leads to increased levels of inequality and social exclusion.
  - ➢ environmental degradation is occurring very rapidly causing problems like land insecurity, excessive air pollution, waste disposal problems.

Technological change on sustainable urbanization

- ❑ Technological change involves the introduction of something new (or) a new idea, method (or) device.
- ❑ A technological innovation, as part of technological change, allows organisations to test new ideas at speeds and prices that were newer anticipated a decade ago.
- ❖ Technological innovation has changed the overall effectiveness and benevolence over time and with regard to sustainability.
- ❖ Upgrading of industrial structure improves the sustainable urbanization.
- ❖ Technological change and sustainability are closely related to each other.
- ❖ Both factors form the innovation in order to improve the effectiveness of environmental and social development and economic progress.

❖ The combination of digital technology in the business model will establish and empower a city to be more sustainable.